BENCHMARKING SINGLE-CELL ANALYSIS

A task-spanning evaluation of analytical quality and monitored computational cost

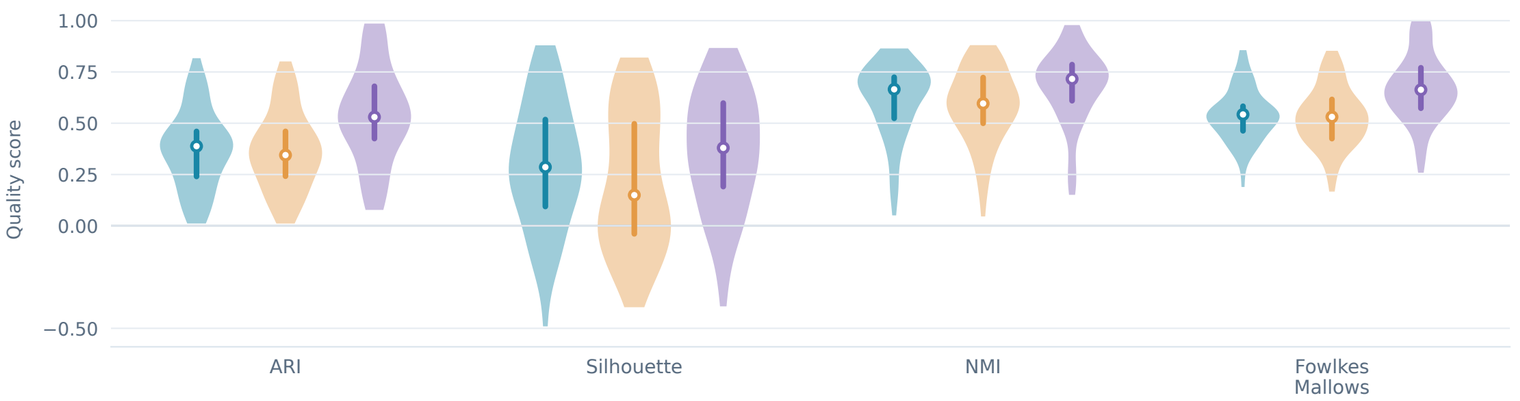
07	WORKFLOW TYPES	72	EVALUATED DATASETS	61	CLUSTERING DATASETS	02	EVALUATION DIMENSIONS

A Benchmark suite and evaluation scope

ANALYSIS TASK	DATASETS	METHODS / SCORING APPROACHES	EVALUATION METRICS
Clustering	61	Scanpy / Seurat / scVI	ARI · NMI · silhouette · Fowlkes–Mallows
Imputation	3	MAGIC / SAVER	MSE · Poisson score*
Multimodal integration	2	MultiVI / MOFA / WNN	Graph connectivity · batch ASW · conservation
Cell type annotation	2	SingleR / CellTypist / scANVI	Accuracy · macro/micro/weighted F1
Cell–cell communication	2	LIANA: 9 scorers × max/sum	Precision–recall AUC · odds ratio
Batch integration	1	Liger / Seurat / Harmony / scVI	Graph connectivity · kBET · batch ASW
Trajectory inference	1	Scanpy	Graph/tree edit · kernel · Jaccard

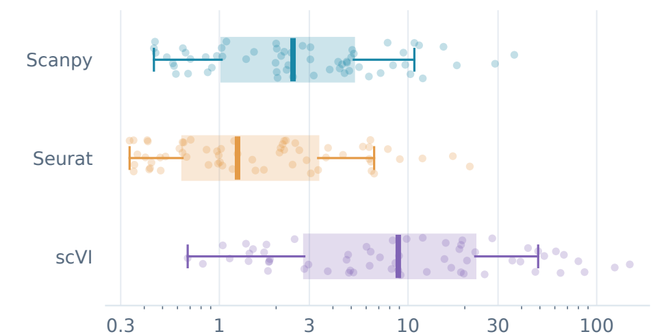
B Clustering performance across 61 datasets

Each violin contains one score per dataset; center bars mark medians



C Execution-window duration

Computed from first and last recorded monitor timestamps



MEDIAN Scanpy 2.46m · Seurat 1.25m · scVI 8.88m

D Integration benchmarks

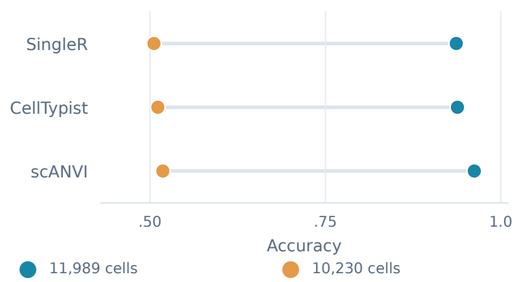
Scores from three datasets (separate single- and multimodal tasks)

	Graph connection	Batch-label ASW	Bio. conservation
PBMC / Liger	**0.93**	**0.90**	0.59
PBMC / Seurat	**0.93**	**0.91**	0.40
PBMC / Harmony	**0.93**	**0.92**	0.31
PBMC / scVI	**0.96**	**0.91**	0.58
Cell line / MultiVI	**1.00**	**0.93**	0.24
Cell line / MOFA	**0.88**	**0.95**	0.21
Cell line / WNN	**0.93**	**0.93**	0.20
Kidney / MultiVI	0.75	**0.93**	0.22
Kidney / MOFA	0.59	**0.91**	0.17
Kidney / WNN	0.72	**0.88**	0.22

Scores 0–1 • Darker tiles = higher recorded values

E Cell-type annotation

Accuracy across two PBMC datasets



F Imputation error

MSE across three datasets (lower is better)

